

46 YEARS OF EXCELLENCE

NARAYANA
IIT - JEE ACADEMY - INDIA

Sec: SR.IIT _*CO-SC(MODEL-A&APEX) **Date: 15-02-26**
Time: 3 Hrs **Max. Marks: 180**

15-02-26_SR.IIT_STAR CO-SC(MODEL-A&APEX)_JEE ADV_RTA-2_SYLLABUS

MATHEMATICS: **PRESENT WEEK (100%):** Indefinite & Definite Integrals, Area under Curve, Differential equation

PHYSICS: **PRESENT WEEK (100%):** Ray optics, Wave optics, EM Waves, Measuring Instruments, Current Electricity

CHEMISTRY: **PRESENT WEEK (100%):** Group-16, Chemical Kinetics, Liquid solution, Alcohol, Phenol, Ether, Group-17

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TIME:3hrs **IMPORTANT INSTRUCTIONS** **Max Marks: 180**

MATHEMATICS

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 1 – 3)	Questions with **Multiple Correct** Choice (partial marking scheme) (+1,0)	+4	-2	3	12
Sec – II (Q.N : 4 – 7)	Questions with **Single Correct** Options	+3	-1	4	12
Sec – III (Q.N : 8 – 13)	Questions with **Non-Negative Integer** type	+4	0	6	24
Sec – IV (Q.N : 14 – 17)	Questions with **MATRIX MATCH**	+3	-1	4	12
Total				**17**	**60**

PHYSICS

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 18 – 20)	Questions with **Multiple Correct** Choice (partial marking scheme) (+1,0)	+4	-2	3	12
Sec – II (Q.N : 21 – 24)	Questions with **Single Correct** Options	+3	-1	4	12
Sec – III (Q.N : 25 – 30)	Questions with **Non-Negative Integer** type	+4	0	6	24
Sec – IV (Q.N : 31 – 34)	Questions with **MATRIX MATCH**	+3	-1	4	12
Total				**17**	**60**

CHEMISTRY

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 35 – 37)	Questions with **Multiple Correct** Choice (partial marking scheme) (+1,0)	+4	-2	3	12
Sec – II (Q.N : 38 – 41)	Questions with **Single Correct** Options	+3	-1	4	12
Sec – III (Q.N : 42 – 47)	Questions with **Non-Negative Integer** type	+4	0	6	24
Sec – IV (Q.N : 48 –51)	Questions with **MATRIX MATCH**	+3	-1	4	12
Total				**17**	**60**

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MATHEMATICS

MAX.MARKS: 60

SECTION - I (Maximum Marks : 12)

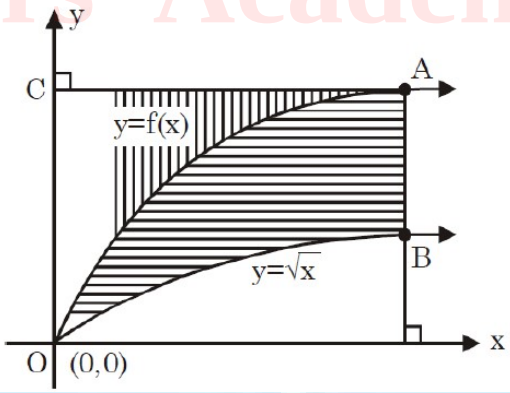
This section contains **THREE (03)** questions.
Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
For each question, choose the correct option(s) to answer the question.
Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks: +3 If all the four options are correct but ONLY three options are chosen.
Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks: -2 In all other cases.

1. If $\int \frac{x^4+9}{x^6\left(x^2-9\right)}dx = \frac{A}{243}\ell n\left|\frac{x-3}{x+3}\right| + \frac{10}{B}\frac{1}{x} + \frac{1}{Cx^3} + \frac{1}{5x^5} + k$ (where k is integration constant). Then

A) $A=5$ B) $\sqrt{B}=3$ C) $(C)^{\frac{1}{3}}=3$ D) $\sqrt{B}=9$

2. Let f(x) is continuous function as shown in the figure. If the area bounded by the curve $y=f(x), y=\sqrt{x}$ and line segment AB is equal to area bounded by $y=f(x), y-$axis and line segment AC then which of the following is (are) correct ?



A) If $\int_0^1 f(x)dx = \frac{1}{3}$ then $f\left(\frac{1}{4}\right) = \frac{1}{2}$

B) If $\int_0^1 f(x)dx = \frac{4}{3}$ then $f\left(\frac{1}{4}\right) = 1$

C) There is only one such function 'f' exist.

D) There are infinitely many such function(s) 'f' exist.

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3. Let $f:R-\{0\}\to R$ be a function which is differentiable in its domain and satisfies the equation $f(x+y)=f(x)+f(y)+\frac{x+y}{xy}-\frac{1}{x+y}$ for all possible x, y $\in$ R. Also, if $f'(1)=2$, then which of the following option(s) is (are) correct?

A) The value of $\int_1^e f(x)\,dx$ is equal to $\frac{e^2-1}{2}$

B) The number of solution(s) of the equation $|f(|x|)|=|x|$ is equal to 2

C) There are 3 integers in the range of possible values of $\frac{4f(x)}{x^5-\frac{1}{x}}$

D) There are 2 integers in the range of possible values of $h(x)=\frac{3f(x)}{x^5-\frac{1}{x}}$

SECTION–II (Maximum Marks: 12)

This section contains **FOUR (04)** questions.
Each question has FOUR options for correct answer(s). **ONLY ONE** of these four option is the correct answer.
For each question, choose the correct option corresponding to the correct answer.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If only the correct option is chosen.
Zero Marks: 0 If none of the option is chosen.(i.e the question is un answered)
Negative Marks: -1 In all other cases.

4. The differential equation $\{xy^3(1+\cos x)-y\}dx+xdy=0$ represents the curve $\frac{x^2}{ay^2}=\frac{x^3}{b}+x^2\sin x+\cos x-d\sin x+k$ Then $a+b+c+d$ is equal to (where $k\in R$ is an independent arbitrary constant)

A) 8 B) 9 C) 10 D) 11

5. Area bounded by $y=\{x\}^{[x]}+[x]$, x-axis between x = 0 and x = 3 (where [.] denotes greatest integer function and {.} denotes fractional part function) is

A) 3 B) $\frac{9}{2}$ C) $\frac{29}{6}$ D) $\frac{25}{6}$

6. The following sum of series of infinite terms is equal to :

$\frac{1}{4}-\frac{1}{5}+\frac{1}{9}-\frac{1}{10}+\frac{1}{14}-\frac{1}{15}+\frac{1}{19}-\frac{1}{20}+....$

A) $\int_0^1\frac{x^2dx}{1+x+x^2+x^3}$ B) $\int_0^1\frac{xdx}{1+x+x^2}$

C) $\int_0^1\frac{x^3dx}{1+x+x^2+x^3+x^4}$ D) $\int_0^1\frac{x^4dx}{1+x+x^2+x^3+x^4+x^5}$

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7. Let x_1 and x_2 $(x_1 < x_2)$ be two values of x satisfying $\left||x|^2 - 4|x| + 5\right| + \left|x^3 - 4x\right| + 3\Big| = 4$ and $f : R \to R$ be a continuous function satisfying

$f(x) - f\left(x + \frac{1}{4}\right) + f\left(x + \frac{1}{2}\right) - f\left(x + \frac{3}{4}\right) = 0 \forall x \in R$ .If $\int_{\frac{1}{2}}^{\frac{3}{2}} f(x)dx = 6$, then value of $\int_{x_1}^{x_2} f(x)dx$ equals

A) 30 B) 18 C) 24 D) 12

SECTION-III (Maximum Marks : 24)

This section contains **SIX (06)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**
For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +4 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.

8. Let $y = y(x)$ be the solution of the differential equation $\frac{dy}{dx} = \frac{4y^3 + 2yx^2}{3xy^2 + x^3}, y(1) = 1$. If for some $n \in N$, $y(2) \in [n-1, n]$, then n is equal to

9. If $\frac{1}{e-1}\sum_{n=1}^{73} \frac{1}{(n+1)!}\int_{-n}^{n} e^{\{|x|\}} \frac{dx}{1+e^{73x}} = 1 - \frac{1}{m!}$, then the value of m is ____

(Where {.} denotes Fractional Part Function)

10. If $\int \left(\sqrt[6]{\frac{x}{x-2}} - \sqrt[4]{\frac{x}{x-2}}\right)\frac{dx}{x^2 - 2x} = f(x)$ and $f(0) = 0$ and if $f(3) = A.\sqrt[4]{3} - B\sqrt[6]{3}$ (where $A, B \in N$), then $(A + B)$ equals

11. The value of the definite integral

$$\frac{4}{\pi}\int_0^{\frac{\pi}{2}} \frac{\cos^4 x + \sin x \cos^3 x + \sin^2 x \cos^2 x + \sin^3 x \cos x}{\sin^4 x + \cos^4 x + 2\sin x \cos^3 x + 2\sin^2 x \cos^2 x + 2\sin^3 x \cos x} dx$$

12. If $f(x) = \int \frac{1}{x}\left(\left\{\frac{x}{4x^2 - 2x + 9}\right\} + \left\{-\frac{x}{9x^2 - 2x + 4}\right\}\right)dx$, $f(5) - f\left(\frac{1}{5}\right) = \ell nq + 3\tan^{-1} r$

(where q, r $\in$ Z and {.} denotes fractional part function), then the value of (q + r) is equal to

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13. Suppose $f:R\to R^+$ and $g:R\to R^+$ are differentiable functions such that $x\cdot g(f(x)).f'(g(x)).g'(x)=f(g(x)).g'(f(x)).f'(x)$ for all real x. Further $\int_0^a f(g(x))dx=1-\frac{e^{-2a}}{2}$ for all $a\in R$. Given that $g(f(0))=1$.

If $p(x)=\int\frac{(x-\ell n2)e^{x+\ell n2}}{e^{\ell n(1+e^x)^2}}dx$, then the value of $9e^{P(\ell n2)-P(0)}-\ell n(g(f(4)))$

SECTION-IV
(Maximum Marks : 12)

This section contains **FOUR (04)** Matching List Sets. Each set has ONE Multiple Choice Question
Each set has TWO lists: List-I and List-II. List-I has Four entries (P), (Q), (R) and (S) and
List-II has Five entries (1), (2), (3), (4) and (5).
FOUR options are given in each Multiple Choice Question based on List-I and List-II and ONLY ONE of these four options satisfies the condition asked in the Multiple Choice Question.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +3ONLYif the option corresponding to the correct combination is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: – 1 in all other cases.

14. Match List-I with List-II and select the correct answer using the code given below the list.

	List-I		List-II
P)	If the order of the differential equation $\left[1+\left(\frac{dy}{dx}\right)^2\right]^{3/2}=\lambda\frac{d^2y}{dx^2},\lambda\neq 0$ is β, then $\beta=$	1)	2
Q)	The order of the differential equation $y=\tan\left(\frac{\pi}{4}+ax\right)\tan\left(\frac{\pi}{4}-ax\right)+ce^{bx+d}$ is $(a,b,c,d$ are parameters$)$	2)	3
R)	The value of $\lim_{x\to\infty}y(x)$ obtained from the differential equation $\frac{dy}{dx}=y-y^2$ where $y(0)=2$ is	3)	4
S)	The order and degree of differential equation of all tangent lines to parabola $x^2=4y$ are α and β respectively then $(\alpha+\beta)$ equals	4)	1
		5)	0

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A) $P \to 2; Q \to 1; R \to 4; S \to 3$ B) $P \to 3; Q \to 1; R \to 2; S \to 4$

C) $P \to 1; Q \to 1; R \to 4; S \to 2$ D) $P \to 4; Q \to 1; R \to 3; S \to 2$

15. Match the following differential equations in List-I with the general solutions in List-II.

	List-I		List-II
P)	$(xy + 2x^2y^2)ydx + (xy - x^2y^2)xdy = 0$	1)	$xy + \ell n\left\|\frac{x}{y}\right\| = \frac{1}{xy} + c$
Q)	$(x^2y^2 + xy + 1)ydx + (x^2y^2 - xy + 1)xdy = 0$	2)	$\ell n\left\|\frac{x}{y}\right\| = xy + c$
R)	$(x^2 + y^2 + 2x)dx + 2ydy = 0$	3)	$x + \ell n(x^2 + y^2) = c$
S)	$y(1 - xy)dx = x(1 + xy)dy$	4)	$2\ell n\|x\| = \ell n\|y\| + \frac{1}{xy} + c$
		5)	$\ell n\|x\| = \ell n\|y\| + \frac{1}{xy} + c$

(where 'c' is an independent arbitrary constant)

A) $P \to 4; Q \to 2; R \to 1; S \to 3$ B) $P \to 4; Q \to 1; R \to 3; S \to 2$

C) $P \to 1; Q \to 2; R \to 3; S \to 4$ D) $P \to 3; Q \to 1; R \to 2; S \to 4$

16. Match List-I with List-II and select the correct answer using the code given below the list.

	List-I		List_II
P)	Solution of differential equation $y(2x^4 + y)\frac{dy}{dx} = (1 - 4xy^2)x^2$ is	1)	$y^2 = (\ell nx)^2 + C$
Q)	Solution of differential equation $\frac{\sum_{n=1}^{\infty} \frac{x^{2n-1}}{(2n-1)!}}{\sum_{n=1}^{\infty} \frac{x^{2n-2}}{(2n-2)!}} = \frac{dx - dy}{dx + dy}, n \in N$ is	2)	$3(x^2y)^2 + y^3 - x^3 = C$

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R)	Solution of differential equation $x=\sum_{r=0}^{\infty}\frac{(xy)^r}{r!}\left(\frac{dy}{dx}\right)^r$ is	3)	$e^{x^2}\left(y^2-1\right)+e^{y^2}=C$
S)	Solution of differential equation $\left(e^{x^2}+e^{y^2}\right)y\frac{dy}{dx}+e^{x^2}\left(xy^2-x\right)=0$	4)	$2ye^{2x}=ce^{2x}-1$
		5)	$y.e^{2x}=ce^{2x}-1$

(where 'C' is an independent arbitrary constant)

A) $P\to 2;Q\to 1;R\to 4;S\to 3$ B) $P\to 1;Q\to 4;R\to 2;S\to 3$

C) $P\to 2;Q\to 4;R\to 1;S\to 3$ D) $P\to 3;Q\to 2;R\to 4;S\to 1$

17. Match the statement in List-I with given value in List-II.

List-I		List-II	
P)	If area bounded by $x=(\theta-\sin\theta);y=(1-\cos\theta);0\le\theta\le 2\pi$ is $\lambda\pi$; then the value of λ can be equal to	1)	6
Q)	If the least area of triangle formed by tangent to the circle $x^2+y^2=1$ and $x=0,y=0$ is A, then A is coprime with	2)	3
R)	Area of triangle formed by any tangent to the curve $y=\frac{2x^2-x-1}{2x^2+5x+2}\left(x\neq -2,-\frac{1}{2}\right)$ and pair of lines $xy-x+2y=2$ is	3)	80
S)	If the maximum area bounded by the curves $x^2=4ay, y=ax, y=\frac{x}{a}, 1\le a\le 2$ is α, then value of $(\alpha+a)$ is	4)	84
		5)	86

A) $P\to 2;Q\to 1,2;R\to 1,4,5;S\to 1,2$ B) $P\to 2;Q\to 1,2,3,4,5;R\to 1;S\to 4$

C) $P\to 2;Q\to 1,2,3,4,5;R\to 2,5;S\to 1,4$ D) $P\to 3;Q\to 2;R\to 4;S\to 4$

For More Join: @IITJEE_Advanced

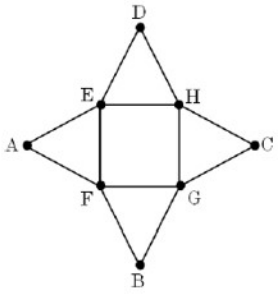
PHYSICS **MAX.MARKS: 60**

SECTION - I (Maximum Marks : 12)

This section contains **THREE (03)** questions.
Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
For each question, choose the correct option(s) to answer the question.
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Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks: -2 In all other cases.

18. The resistor network in figure is built from twelve identical conductors with each conductor having a resistance $R = 0.50\Omega$, then choose the **correct** option(s)



A) The equivalent resistance between AC is $R_{AC} = \frac{2}{3}\Omega$

B) The equivalent resistance between EH is $R_{EH} = \frac{1}{4}\Omega$

C) The equivalent resistance between AE is $R_{AE} = \frac{5}{16}\Omega$

D) The equivalent resistance between AE is $R_{AE} = \frac{5}{8}\Omega$

19. In ordinary Vernier calpers, 10th division of the Vernier scale coincides with 9th division of the main scale. In a specially designed Vernier calipers the Vernier scale is so constructed that 10th division on it coincides with 11th division on the main scale. Each division on the main scale equal to 1 mm. The calipers have a zero error as shown in the figure-1. When the Vernier caliper is used to measure a length, the concerned portion of its scale is shown in figure-II. Then choose the **correct** option(s)

Figure-I

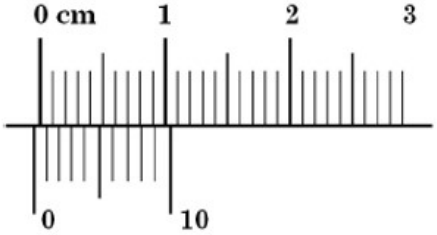
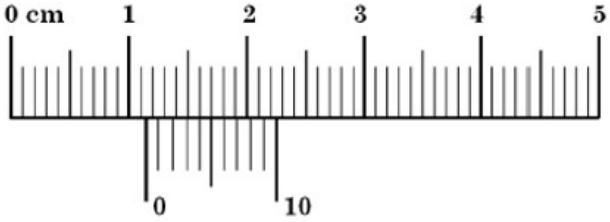


Figure-II



For More Join: @IITJEE_Advanced

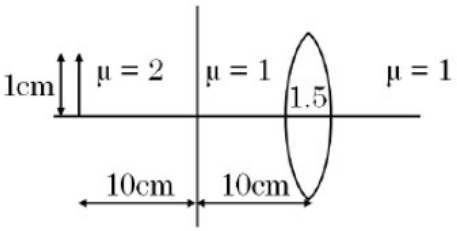
A) Zero error in the calipers has magnitude 0.7 mm.

B) The length being measured is 1.08 cm

C) The length being measured is 1.22 cm

D) The given Vernier scale follows the principle of Vernier scale, yet can't be used satisfactorily.

20. An object of length 1 cm is placed on principal axis of an equi-convex lens of radius 5 cm. Distance between the lens and object is 20 cm. Space between the lens and object is filled with medium of two different refractive index 2 and 1 as shown in the figure. Boundary of both medium is mid-way between the object and lens as shown in figure, (consider all rays are paraxial). Then choose the **correct** option(s)



A) The image will be formed at distance of 7.5 cm from the optical centre

B) The image will be formed at distance of 10 cm from the optical centre

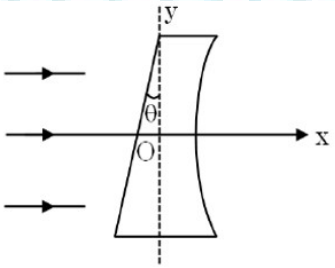
C) The size of the image is 0.5 cm

D) The size of the image is 0.4 cm

SECTION–II (Maximum Marks: 12)

This section contains **FOUR (04)** questions.
Each question has FOUR options for correct answer(s). **ONLY ONE** of these four option is the correct answer.
For each question, choose the correct option corresponding to the correct answer.
Answer to each question will be evaluated according to the following marking scheme:
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Zero Marks: 0 If none of the option is chosen.(i.e the question is un answered)
Negative Marks: -1 In all other cases.

21. Due to manufacturing defect, the plane surface of a thin plano-concave lens has been made tilted at a small angle θ outwards from its usual place. The spherical surface has radius of curvature R and refractive index of the material of the lens is μ. A parallel beam of light is incident as shown. In the co-ordinate system shown find co-ordinate of the point where the rays will focus or appear to be diverging from.



A) $\frac{-R}{\mu-1},-R\theta$ B) $\frac{-R}{\mu-1},+R\theta$ C) $\frac{-2R}{\mu-1},-R\theta$ D) $\frac{-2R}{\mu-1},+R\theta$

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22. An electromagnetic wave has the form

$$E = \hat{i}E_0 \sin\left(\omega t + \frac{\omega z}{c}\right) + \hat{j}2E_0 \sin\left(\omega t + \frac{\omega z}{c}\right)$$

The formula for the magnetic field of this wave as a function of space and time, is

A) $\vec{B} = \frac{2E_0}{c}\sin\left(\omega t + \frac{\omega z}{c}\right)\hat{i} - \frac{E_0}{c}\left(\omega t + \frac{\omega z}{c}\right)\hat{j}$

B) $\vec{B} = \frac{E_0}{c}\sin\left(\omega t + \frac{\omega z}{c}\right)\hat{i} - \frac{2E_0}{c}\sin\left(\omega t + \frac{\omega z}{c}\right)\hat{j}$

C) $\vec{B} = -\frac{2E_0}{c}\sin\left(\omega t + \frac{\omega z}{c}\right)\hat{i} - \frac{E_0}{c}\sin\left(\omega t + \frac{\omega z}{c}\right)\hat{j}$

D) $\vec{B} = -\frac{2E_0}{c}\sin\left(\omega t + \frac{\omega z}{c}\right)\hat{i} + \frac{E_0}{c}\sin\left(\omega t + \frac{\omega z}{c}\right)\hat{k}$

23. A group of students is given two converging lenses. Lens A has a focal length of 12.5 cm, and lens B has a focal length of 50.0 cm. The diameter of each lens is 6.10 cm. Assume wavelength of 500 nm.

Select the correct statement

A) If they want to make a telescope, lens A should be the objective and lens B should be the eyepiece. And if they want to make a microscope, lens B should be the objective and lens A should be the eyepiece.

B) If they want to make a microscope, lens A should be the objective and lens B should be the eyepiece. And if they want to make a telescope, lens B should be the objective and lens A should be the eyepiece.

C) Lens A should be the objective and lens B should be the eyepiece whether they make telescope or microscope.

D) Lens B should be the objective and lens A should be the eyepiece whether they make telescope or microscope.

For More Join: @IITJEE_Advanced

24. Twelve identical resistors each of resistance R, are connected in the configuration of a regular octahedron (a solid made of eight equilateral triangles four of which meet at each vertex). Find the equivalent resistor between any two opposite vertices.

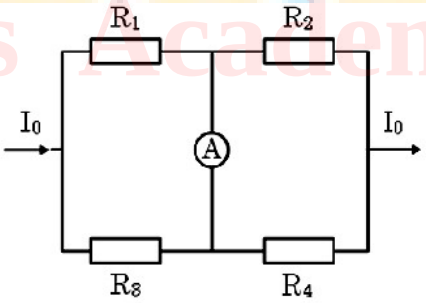


A) $\frac{R}{6}$ B) $\frac{R}{2}$ C) $\frac{2R}{3}$ D) $\frac{6R}{5}$

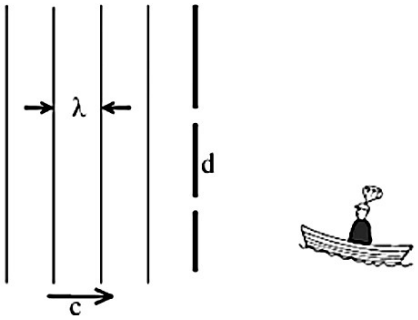
SECTION-III (Maximum Marks : 24)

This section contains **SIX (06)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**
For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +4 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.

25. A current of I A will flow through the ideal ammeter in the electric circuit shown in the figure, and $R_1 = r; R_2 = 2r; R_3 = 3r; R_4 = 4r; I_0 = 1A$? The value of 24 I is



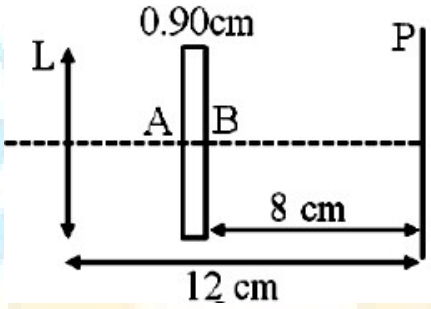
26. A man in a small dinghy approaches from a great distance the wave-breaker dam, in which there are two holes 5m apart. Waves of wavelength $\lambda = 6$ m hit the other side of the dam. The direction of path (in degree) of dinghy from the normal to the dam at which the dinghy is to be rocked the least is



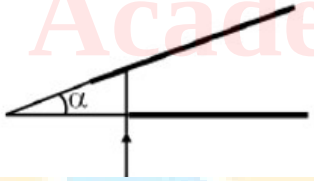
For More Join: @IITJEE_Advanced

27. The object at 2.4 meters in front of the camera lens L is clearly imaged on the photographic film P 12.0 cm behind the lens, and now a glass plate with a refractive index of 1.50, thickness AB = 0.90 cm, parallel to both sides is inserted between the lens and the film, perpendicular to the optical axis, the position is as shown in the figure, and the camera lens can be regarded as a simple thin convex lens. If the distance between the lens, glass plate, and film is kept unchanged, and the object is still clearly imaged on the film, by what distance (in cm) must be object be moved?

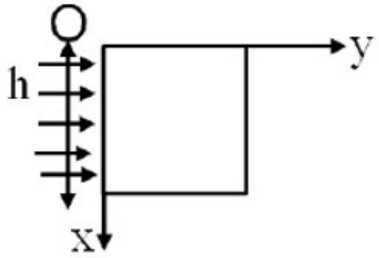
(round off to closest integer)



28. Two very long mirrors are arranged in such a way that the angle of α between their planes can change, taking the values: $8° \le \alpha \le 12°$. The beam of the laser pointer is sent to the upper mirror in the direction perpendicular to the lower mirror. Find the maximum number of reflections.



29. A plane slab as shown has a variable refractive index such that $n = \dfrac{3}{2-\dfrac{x}{h}}$, where x is the distance from the y-axis as shown. h is the thickness of the slab. A beam is incident on it normally. It is seen to follow a trajectory inside the slab whose radius of curvature is ah at x = 0. Find a



30. Plane beam of electromagnetic radiation of frequency 1.25×10^{25} Hz is diffracted by Li^{8} nucleus. The nucleus is totally radiation absorbent and thus is a perfect obstacle. At what angle (in 10^{-3} rad) will the first diffraction minimum lie?(Take R_0 = 1.22 fm)

For More Join: @IITJEE_Advanced

SECTION-IV
(Maximum Marks : 12)

This section contains **FOUR (04)** Matching List Sets. Each set has ONE Multiple Choice Question
Each set has TWO lists: List-I and List-II. List-I has Four entries (P), (Q), (R) and (S) and
List-II has Five entries (1), (2), (3), (4) and (5).
FOUR options are given in each Multiple Choice Question based on List-I and List-II and ONLY ONE of these four options satisfies the condition asked in the Multiple Choice Question.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +3ONLYif the option corresponding to the correct combination is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: – 1 in all other cases.

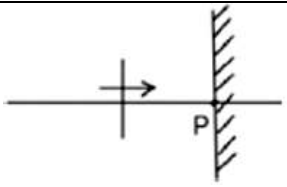
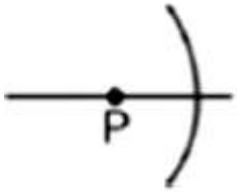
31. List-I contains x and y-components of the electric field and List-II contains the type of polarization of light.

List-I		List-II	
P)	$E_x = \frac{E_0}{\sqrt{2}}\cos(\omega t + kz)$ $E_y = E_0 \sin(\omega t + kz)$	1)	Linearly Polarized
Q)	$E_x = E_0 \sin(\omega t + kz)$ $E_y = E_0 \cos(\omega t + kz)$	2)	Circularly Polarized
R)	$E_x = E_1 \sin(\omega t + kz)$ $E_y = E_2 \sin(\omega t + kz)$	3)	Unpolarized
S)	$E_x = E_0 \sin(\omega t + kz)$ $E_y = E_0 \sin\left(\omega t + kz + \frac{\pi}{4}\right)$	4)	Elliptically Polarized
		5)	None of these

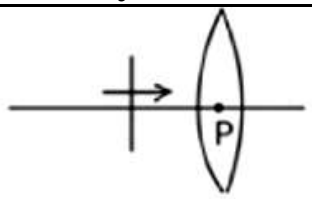
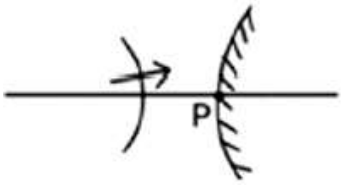
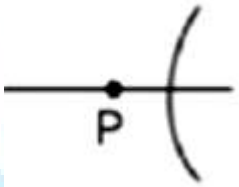
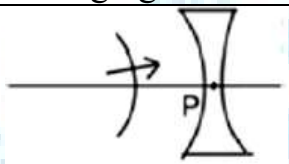
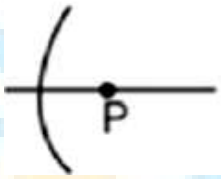
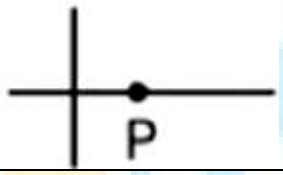
A) $P \to 4; Q \to 2; R \to 4; S \to 1$ B) $P \to 1; Q \to 3; R \to 1; S \to 4$
C) $P \to 4; Q \to 2; R \to 1; S \to 4$ D) $P \to 3; Q \to 1; R \to 3; S \to 2$

32. List-I shows some mirrors or lens with a small wavefront symmetric about the principal axis heading towards them.

List-II shows the possible shape of the wavefront just after being incident on to them. Choose the correct option.

List-I		List-II	
P)	 Plane mirror with a plane wavefront parallel to mirror	1)	

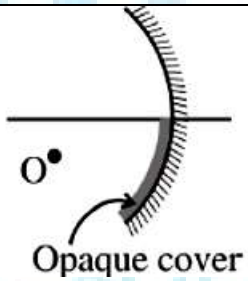
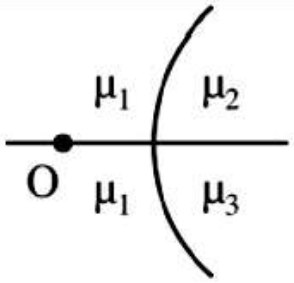
For More Join: @IITJEE_Advanced

Q)	 Converging lens with a plane wavefront perpendicular to principal axis	2)	
R)	 Diverging mirror	3)	
S)	 Diverging lens	4)	
		5)	

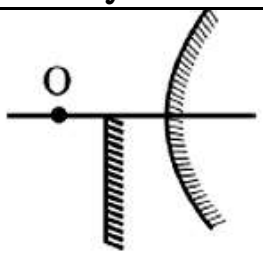
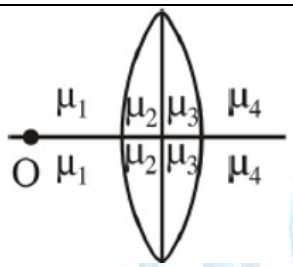
A) $P \rightarrow 2; Q \rightarrow 3; R \rightarrow 5; S \rightarrow 4$ B) $P \rightarrow 2; Q \rightarrow 1; R \rightarrow 3; S \rightarrow 4$

C) $P \rightarrow 5; Q \rightarrow 1; R \rightarrow 4; S \rightarrow 2$ D) $P \rightarrow 5; Q \rightarrow 3; R \rightarrow 4; S \rightarrow 1$

33. Select correct answer using the codes given below:

List-I		List-II	
P)	 Opaque cover	1)	Number of images formed is one
Q)		2)	Number of images formed is two

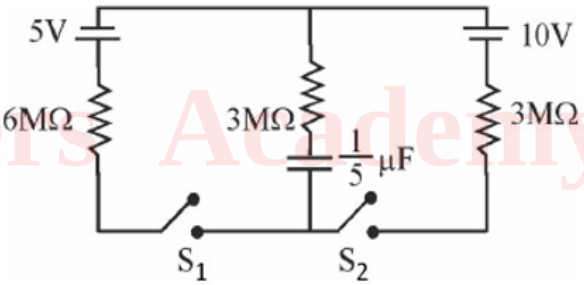
For More Join: @IITJEE_Advanced

R)		3)	Number of images formed is three
S)		4)	Number of images formed is four
		5)	No image is formed

A) $P \to 5; Q \to 2; R \to 4; S \to 3$ B) $P \to 5; Q \to 2; R \to 2; S \to 3$

C) $P \to 1; Q \to 2; R \to 2; S \to 4$ D) $P \to 1; Q \to 2; R \to 2; S \to 1$

34. Three resistors of resistance $6M\Omega, 3M\Omega$ and $3M\Omega$, one capacitor of capacitance $\frac{1}{5}\mu F$, two cells of potential difference 5V and 10V and two switches are connected in the circuit as shown in figure. Both switches are closed simultaneously at t = 0, $\left(e^{-1} = 0.37\right)$



List-I describe the charges and current in the circuit at a particular time and List-II gives possible values of charges and current. What one of the following option is **correct**?

	LIST-I		LIST-II
P)	The maximum charge on the capacitor in μC	1)	0.6
Q)	Charge on capacitor at t = 1 sec in μC	2)	1.54
R)	Current in the branch of capacitor at t = 1 sec in μA	3)	2.06
S)	Current passing through battery of 5V at t = 1 sec in μA	4)	1.00
		5)	0.37

A) $P \to 1; Q \to 5; R \to 3; S \to 4$ B) $P \to 4; Q \to 1; R \to 2; S \to 3$

C) $P \to 1; Q \to 3; R \to 5; S \to 2$ D) $P \to 4; Q \to 1; R \to 5; S \to 2$

For More Join: @IITJEE_Advanced

CHEMISTRY **MAX.MARKS: 60**

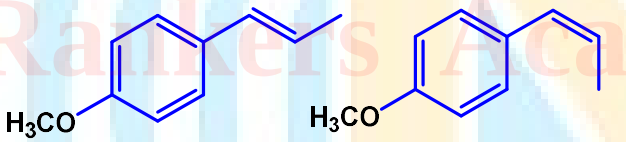
SECTION - I (Maximum Marks : 12)

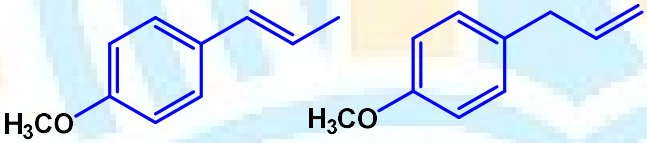
This section contains **THREE (03)** questions.
Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
For each question, choose the correct option(s) to answer the question.
Answer to each question will be evaluated according to the following marking scheme:

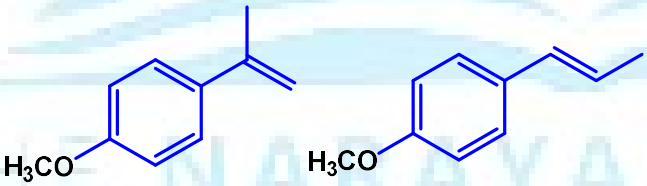
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks: +3 If all the four options are correct but ONLY three options are chosen.
Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks: -2 In all other cases.

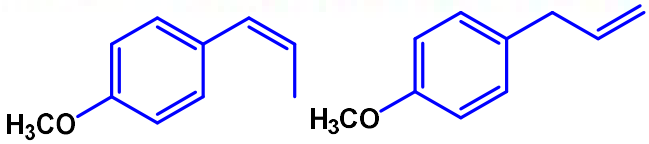
35. Two isomeric compounds, [P] and [Q], are isolated from oil of bay leaf; both are found to have the formula $C_{10}H_{12}O$. Both are insoluble in water, dilute acid, and dilute base. Both give positive tests with dilute $KMnO_4$ and Br_2/CCl_4. Upon vigorous oxidation, both yield anisic acid, P-$CH_3OC_6H_4COOH$. Catalytic hydrogenation of [P] and [Q] gives same compound.

Structure of [P] and [Q] respectively is/are

A) 

B) 

C) 

D) 

For More Join: @IITJEE_Advanced

36. Which of the following has higher value for H_2O_2 as compared to H_2O

A) Melting point, Boiling point & Dielectric constant

B) Melting point, Boiling point and extent of Hydrogen bonding

C) Density, Boiling point and Oxidizing nature

D) Dielectric constant, Viscosity & Density

37. One litre of 2 molar solution of a complex salt $CoCl_3.4NH_3$ (Mol. wt = 233.5) shows an osmotic pressure of 98.40 atm at 300 K. The solution is now treated with 2 litre of 1M $AgNO_3$. Which of the following is/are **correct**?

[Ag = 108] (R = 0.082 L–atm/K–mol)

A) Weight of AgCl preripitated is 287g.

B) The final clear solution will show an osmotic pressure of 98.40 atm.

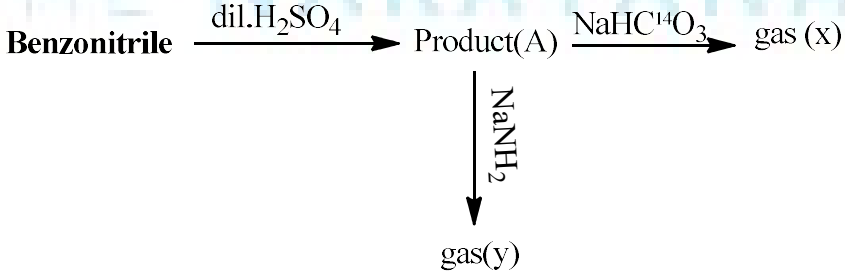
C) The final clear solution will shown an osmotic pressure of 32.80 atm.

D) 2 mole of $\left[Co(NH_3)_4Cl_2\right].NO_3$ will be present in the final solution after addition of $AgNO_3$.

SECTION–II (Maximum Marks: 12)

This section contains **FOUR (04)** questions.
Each question has FOUR options for correct answer(s). **ONLY ONE** of these four option is the correct answer.
For each question, choose the correct option corresponding to the correct answer.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If only the correct option is chosen.
Zero Marks: 0 If none of the option is chosen.(i.e the question is un answered)
Negative Marks: -1 In all other cases.

38.



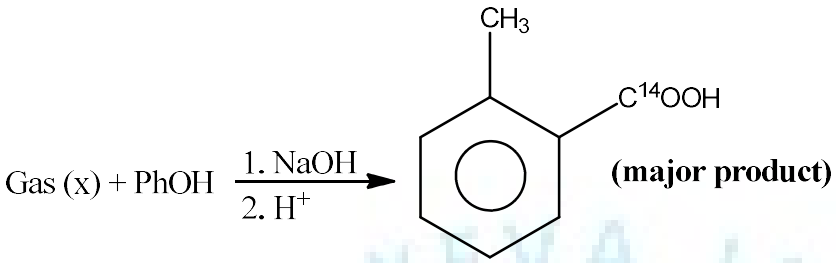
Choose the **CORRECT** option

For More Join: @IITJEE_Advanced

A) Molecular mass of gas (x) is 44.

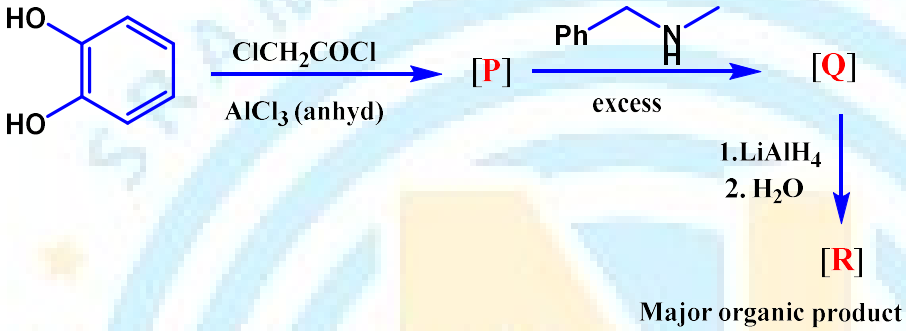
B) Gas (y) is neutral in nature

C) Gas (y) with HCl from white fumes



D)

39. Choose the **incorrect** option(s)



A) [P] is an optically inactive compound

B) [P], [Q] and [R] gives neutral $FeCl_3$ test

C) Difference in molecular weight between [P] and [Q] is 85

D) One of the functional group in [Q] is 3° amide

40. The inversion of cane sugar proceeds with half life of 50 minute at pH = 5 for any concentration of sugar. However at pH = 6, the half life changes to 500 minute for any concentration of sugar. The rate law expression for the inversion of cane sugar is

A) $r = K[\text{sugar}]^1[H^+]^{-1}$ B) $r = K[\text{sugar}]^1[H^+]^2$

C) $r = K[\text{sugar}]^1[H^+]^1$ D) $r = K[\text{sugar}]^1[H^+]^0$

41. For a first order reaction: $A \rightarrow B + C$, all the reactant and product molecules are in gaseous state. If the initial pressure of the reactant molecule is 100 mm Hg and after 30 minutes the total pressure of the mixture containing molecules of all three gases A, B & C is 140 mm Hg, then after how much time the initial pressure of A will be reduced to 50 mm Hg.

$(\log_{10} 3 = 0.5, \log_{10} 2 = 0.3, \log_{10} 5 = 0.7)$

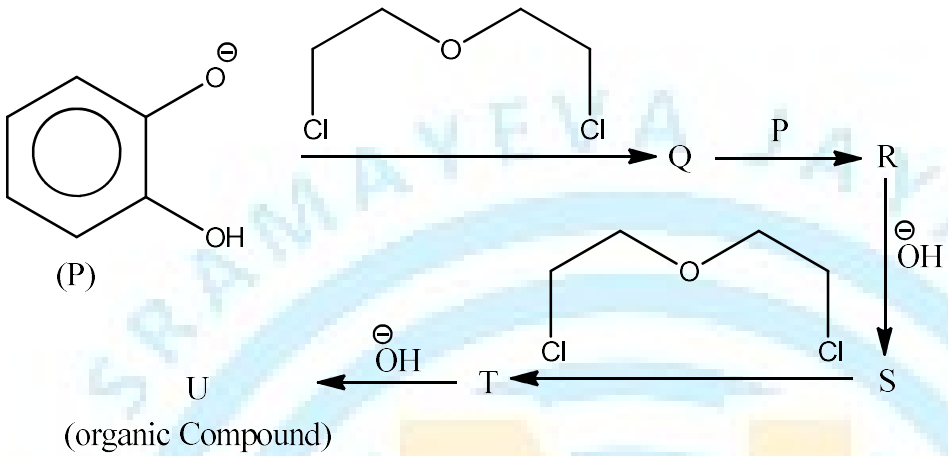
A) 45 min B) 30 min C) 60 min D) 15 min

For More Join: @IITJEE_Advanced

SECTION-III (Maximum Marks : 24)

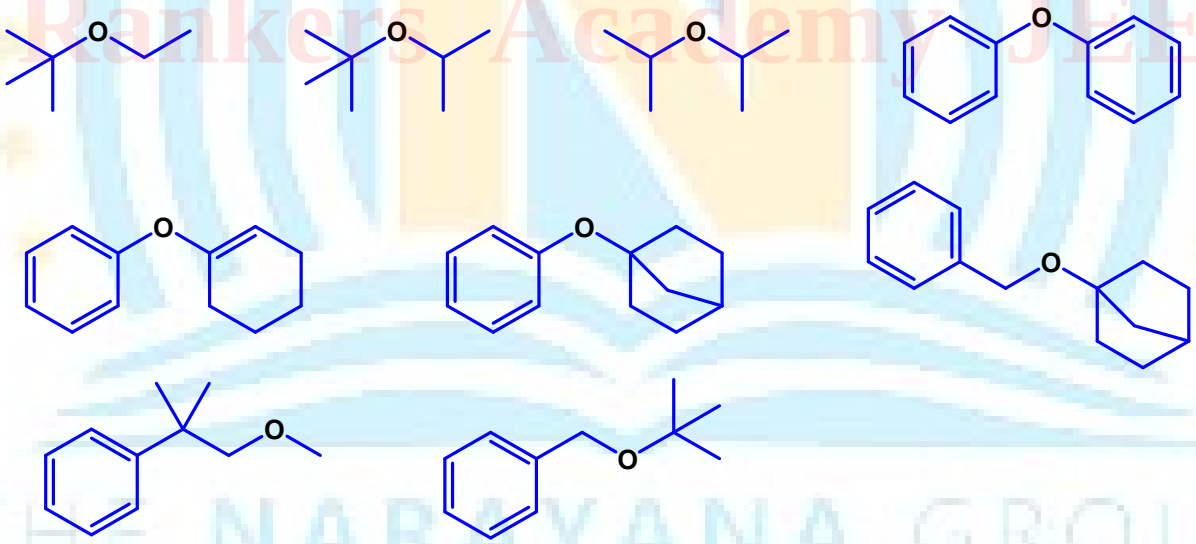
This section contains **SIX (06)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**
For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +4 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.

42.



Number of oxygen atoms in U is _____

43. How many of the following ethers can't be synthesized using Williamson ether synthesis in good yield



44. A very small amount of a non-volatile solute (non-associative, non-dissociative) is dissolved in 100 cm^3 of a solvent. At room temperature, vapour pressure of this solution is 98.7 mm of Hg while that of pure solvent is 100 mm of Hg. If the freezing temperature of this solution is 0.72 K lower than that of pure solvent, what is the value of cryoscopic constant of solvent (in K Kg/mol) ? Round off your answer to the nearest whole number.

Given : Molar mass of solvent = 78 g/mol.

For More Join: @IITJEE_Advanced

45. 0.01 moles of sulphuryl chloride $SO_2Cl_2(g)$ is taken in a sealed tube & heated to 400 K. It decomposes following first order kinetics according to reaction

$$SO_2Cl_2(g) \rightarrow SO_2(g) + Cl_2(g)$$

The tube is broken after 4 hours & gas passed through a 25 ml of an acidified 1 N iodine solution where all SO_2 is oxidised to SO_4^{2-}. The resulting solution required 10 ml of 1 M Hypo solution. What is half-life for decomposition of SO_2Cl_2 in hours.

46. Consider the following orders:

(1) $H_2SO_4 > H_2SO_3$: Boiling point

(2) $H_2O > HF$: Extent of H-bond

(3) $H_2O > H_2O_2$: Strength of H-bond

(4) HF > HCl > HBr: Melting point

(5) para-dichlorobenzene > ortho-dichlorobenzene: Boiling point

(6) Ethylene glycol > Phenol: Viscosity

(7) 1, 3-Dichlorobenzene > 1, 3, 5-Trichlorobenzene: Strength of molecular force

(8) ortho-Hydroxy benzoic acid > para-Hydroxy benzoic acid: Solubility in water.

Then calculate value of $|x^2|$ (where 'x' is total number of correct orders.)

47. How many of the following compounds form HCl on hydrolysis as one of the products (major or minor)

(i) BCl_3 (ii) $BiCl_3$ (iii) SO_2Cl_2 (iv) NCl_3 (v) PCl_5

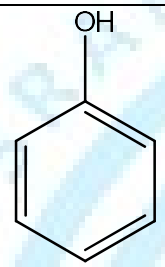
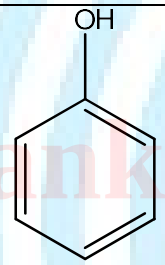
(vi) CrO_2Cl_2 (vii) CH_3Cl (viii) NaCl (ix) $ZnCl_2$

For More Join: @IITJEE_Advanced

SECTION-IV
(Maximum Marks : 12)

This section contains **FOUR (04)** Matching List Sets. Each set has ONE Multiple Choice Question
Each set has TWO lists: List-I and List-II. List-I has Four entries (P), (Q), (R) and (S) and
List-II has Five entries (1), (2), (3), (4) and (5).
FOUR options are given in each Multiple Choice Question based on List-I and List-II and ONLY ONE of these four options satisfies the condition asked in the Multiple Choice Question.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +3ONLYif the option corresponding to the correct combination is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: – 1 in all other cases.

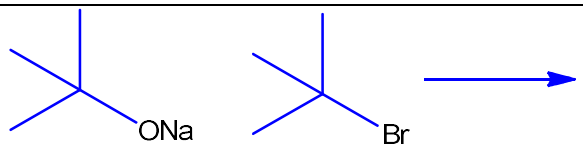
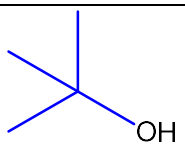
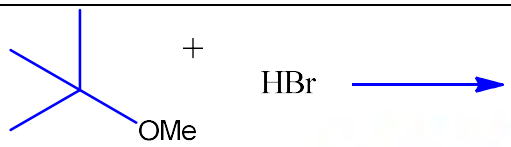
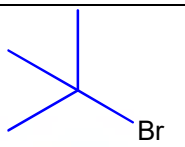
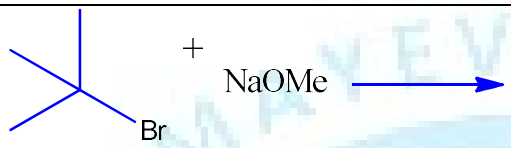
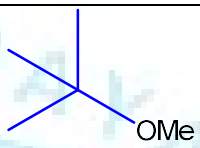
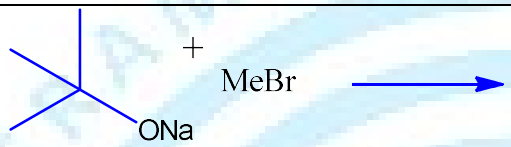
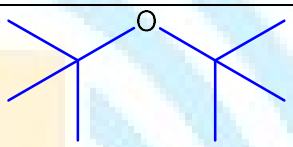
48. Match the following

List-I		List-II	
P)	 i) CO_2, NaOH, high T & P → ii) aq. Workup iii) Ac_2O, pyridine (base)	1)	Methylsalicylate
Q)	 i) CO_2, NaOH, high T & P → ii) aq. Workup iii) CH_3OH, H^+ (cat)	2)	Acetylsalicylic acid
R)	$Ph-CH_2-O-OH \xrightarrow{H^+(catalyst)}$	3)	PhOH + Methanol
S)	$PhOH \xrightarrow[ii)(CH_3)_2SO_4]{i)aq.NaOH}$	4)	$PhOH + CH_2O$
		5)	Methoxy benzene
		6)	1,4-dimethoxy benzene

A) $P \rightarrow 2; Q \rightarrow 1; R \rightarrow 4; S \rightarrow 5$
B) $P \rightarrow 2; Q \rightarrow 1; R \rightarrow 3; S \rightarrow 6$
C) $P \rightarrow 2; Q \rightarrow 1; R \rightarrow 4; S \rightarrow 6$
D) $P \rightarrow 2; Q \rightarrow 1; R \rightarrow 3; S \rightarrow 5$

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49. List-I contains reactions and List-II contains major product(s)

List-I		List-II	
P)		1)	
Q)		2)	
R)		3)	
S)		4)	
		5)	

Match each reaction in List – I with one ore more products in List – II and choose the **correct** option

A) $P \rightarrow 1,5; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 4$ B) $P \rightarrow 1,4; Q \rightarrow 2; R \rightarrow 4; S \rightarrow 3$

C) $P \rightarrow 1,4; Q \rightarrow 1,2; R \rightarrow 3; S \rightarrow 4$ D) $P \rightarrow 4,5; Q \rightarrow 4; R \rightarrow 4; S \rightarrow 3,4$

50. Assuming all the solutes are non volatile and all solutions are ideal and neglect the hydrolysis of cation and anion.

	LIST-I		LIST-II
P)	10 ml 0.1 M NaOH aqueous solution is added to 10 ml 0.1 M HCl aqueous solution	1)	Osmotic pressure of solution increases
Q)	10 ml 0.1 M NaOH aqueous solution is added to 10 ml 0.1 M CH_3COOH aqueous solution	2)	Vapour pressure of solution increases
R)	10 ml 0.1 M HCl aqueous solution is added to 10 ml 0.1 M NH_4OH aqueous solution	3)	Boiling point of solution increases
S)	10 ml 0.1 M HCl aqueous solution is added to 10 ml 0.1 M KOH aqueous solution	4)	Freezing point of solution increases
		5)	Osmotic pressure remains constant

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A) $P \rightarrow 2,4; Q \rightarrow 2,4; R \rightarrow 2,4; S \rightarrow 2,4$

B) $P \rightarrow 1,2; Q \rightarrow 1,2,5; R \rightarrow 2,3; S \rightarrow 3,4$

C) $P \rightarrow 1,2;\ Q \rightarrow 3,4;\ R \rightarrow 3,4,5; S \rightarrow 2,3$

D) $P \rightarrow 2,3; Q \rightarrow 1,2; R \rightarrow 2,3; S \rightarrow 1,3$

51. For the reaction

$2A(g) \rightarrow B(g)$ the initial half life for disappearance of A is 40 seconds starting with 4 moles of A in a 5 litre container at 400 K

List-I		List-II	
P)	If reaction follows first order kinetics, then initial rate of reaction	1)	3.46×10^{-4} M / s
Q)	If reaction follows zero order kinetics, then rate of reaction at time : t = 40 sec.	2)	6.93×10^{-3} M / s
R)	If reaction follows first order kinetics, then rate of reaction at 95% completion.	3)	5×10^{-3} M / s
S)	If reaction follows second order kinetics, then rate of reaction at 99% completion	4)	10^{-6} M / s
		5)	6.25×10^{-5} M / s

The **correct** option is

A) $P \rightarrow 2; Q \rightarrow 3; R \rightarrow 1; S \rightarrow 4$

B) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 4$

C) $P \rightarrow 5; Q \rightarrow 1; R \rightarrow 2; S \rightarrow 4$

D) $P \rightarrow 2; Q \rightarrow 1; R \rightarrow 4; S \rightarrow 3$

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